LAB 7 — Fuzz Testing for Input Validation Bugs

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➔ Before fix

```
PS C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7\before_fix> python self_storage.py
==== Running parse_rental ====
Enter unit:months (e.g. medium:6): medium:6
Unit and Months: ('Medium', 6.0)

==== Running duration_average ====
Enter rental months as CSV (e.g. 3,6,12): 3,6,12
Average Duration: 7

==== Running payable_rent ====
Enter rent|discount (e.g. 2400|200): 2400|200
Payable Rent: 2632.0
```

➔ Hypothesis tests against original application

```
PS C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7\before_fix> python -m pytest -q test_self_storage.py
FFF                                                                      [100%]
============================ FAILURES ============================
_____________________ test_parse_rental_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7\before_fix\test_self_storage.py", line 6, in test_parse_rental_no_crash
  |     def test_parse_rental_no_crash(s):
  |                    ^^^
  |   File "C:\Users\Nehaa\AppData\Roaming\Python\Python314\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7\before_fix\test_self_storage.py", line 8, in test_parse_rental_no_crash
    |     result = parse_rental(s)
    |   File "C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7\before_fix\self_storage.py", line 3, in parse_rental
    |     return unit.strip().title(), float(months)
    |                                  ~~~~~^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7\before_fix\test_self_storage.py", line 10, in test_parse_rental_no_crash
    |     assert False, f"parse_rental crashed with {s!r}: {type(exc).__name__}: {exc}"
    | AssertionError: parse_rental crashed with ':': ValueError: could not convert string to float: ''
    | assert False
    | Failing test case: test_parse_rental_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
```

```
_____ test_duration_average_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7\before_fix\test_self_storage.py", line 14, in test_duration_average_no_crash
  |     def test_duration_average_no_crash(s):
  |                    ^^^
  |   File "C:\Users\Nehaa\AppData\Roaming\Python\Python314\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7\before_fix\test_self_storage.py", line 16, in test_duration_average_no_crash
    |     result = duration_average(s)
    |   File "C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7\before_fix\self_storage.py", line 6, in duration_average
    |     values = [int(r) for r in months_csv.split(',')]
    |                  ~~~^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7\before_fix\test_self_storage.py", line 18, in test_duration_average_no_crash
    |     assert False, f"duration_average crashed with {s!r}: {type(exc).__name__}: {exc}"
    | AssertionError: duration_average crashed with '': ValueError: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_duration_average_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7\before_fix\test_self_storage.py:17
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7\before_fix\test_self_storage.py", line 16, in test_duration_average_no_crash
```

```
_____ test_payable_rent_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7\before_fix\test_self_storage.py", line 22, in test_payable_rent_no_crash
  |     def test_payable_rent_no_crash(s):
  |                    ^^^
  |   File "C:\Users\Nehaa\AppData\Roaming\Python\Python314\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7\before_fix\test_self_storage.py", line 24, in test_payable_rent_no_crash
    |     result = payable_rent(s)
    |   File "C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7\before_fix\self_storage.py", line 11, in payable_rent
    |     total = float(rent) * 1.18 - float(discount)
    |             ~~~~~^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7\before_fix\test_self_storage.py", line 26, in test_payable_rent_no_crash
    |     assert False, f"payable_rent crashed with {s!r}: {type(exc).__name__}: {exc}"
    | AssertionError: payable_rent crashed with '|': ValueError: could not convert string to float: ''
    | assert False
    | Failing test case: test_payable_rent_no_crash(
    |     s='|',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
```

```
========== short test summary info ==========
FAILED test_self_storage.py::test_parse_rental_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exc...
FAILED test_self_storage.py::test_duration_average_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exc...
FAILED test_self_storage.py::test_payable_rent_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exc...
3 failed in 12.89s
```

➔ Fixed application execution

```
PS C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7> python self_storage.py
==== Running parse_rental ====
Enter unit:months (e.g. medium:6): medium:6
Unit and Months: ('Medium', 6.0)

==== Running duration_average ====
Enter rental months as CSV (e.g. 3,6,12): 3,6,12
Average Duration: 7

==== Running payable_rent ====
Enter rent|discount (e.g. 2400|200): 2400|200
Payable Rent: 2632.0
PS C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7>
```

➔ Hypothesis tests against fixed application

```
PS C:\Users\Nehaa\OneDrive\Documents\sem5\SE\LAB7> python -m pytest -q test_self_storage.py
.......                                                                  [100%]
7 passed in 7.28s
```